# AHMED THARWAT MUHAMMAD IBRAHIM

## AI & Data Science | Machine Learning | Data Analysis

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### PROFESSIONAL SUMMARY

Machine Learning Engineer and Data Analyst who builds end-to-end ML pipelines and analysis reports in Python, from data cleaning and EDA to modeling and evaluation. Completed the NTI Machine Learning program (95%) and is now training in the IBM Data Scientist track at DEPI.

### INTERNSHIP

**AI & Data Science Trainee (Data Scientist Track) — DEPI** 2026 – Present

IBM-designed track: Python, SQL, data analysis and visualization, machine learning, MLOps (MLflow, Hugging Face).

**Machine Learning Trainee — NTI** 2026

ML fundamentals, data analysis, predictive modeling and real-world projects in Python. Score: 95%.

### FEATURED PROJECTS

**DrowsyGuard AI — Driver Drowsiness Detection** Team Project · In Progress

- Detects driver drowsiness in real time using ML and computer vision and triggers an alert before fatigue leads to an accident. *Tech: Python, Machine Learning, Computer Vision*

**Bank Term Deposit Prediction — ML Pipeline** Link

- End-to-end pipeline on the UCI Bank Marketing dataset (~45K records); handled class imbalance (~88/12) and data leakage. XGBoost performed best: 82.4% accuracy, 0.60 recall (Random Forest 81.6%, Logistic Regression 76.3%). *Tech: pandas, scikit-learn, XGBoost*

**Sales Analysis Report — Python & Excel** Link

- Cleaned and analyzed 500 e-commerce orders (duplicates, missing values, delivery outliers), built 6 charts and generated an Excel report with insights; Electronics drives 71% of $585K revenue. *Tech: pandas, Matplotlib, openpyxl*

**E-commerce Sales Analysis** Link

- Data cleaning, EDA and business insights on 75 orders (631K EGP), with 6 charts and an Excel report using live formulas. *Tech: pandas, NumPy, Matplotlib, Excel*

**Loan Approval Prediction** Link

- EDA and Logistic Regression on 500 loan applications; Credit Score was the only strong driver (approval rate 45% to 76% across score bands). Model reached 61% accuracy vs a 58.6% baseline. *Tech: scikit-learn, pandas, seaborn*

### EDUCATION

**Zagazig National University — Faculty of Computers and Information** Expected 2028

Bachelor's Degree in Artificial Intelligence & Data Analysis

### SKILLS

**Technical:** Machine Learning, Data Analysis, Data Cleaning, Feature Engineering, Model Evaluation, Data Visualization, End-to-End ML Pipelines, Python, SQL, MLflow, Hugging Face

**Tools:** Python (pandas, NumPy, scikit-learn, XGBoost, Matplotlib), Jupyter, Excel, VS Code, GitHub

**Soft Skills:** Analytical Thinking, Problem-Solving, Communication, Fast Learning

### CERTIFICATIONS

- Machine Learning Training Program — NTI, 2026 · 95%
- AI & Data Science (Data Scientist Track, IBM) — DEPI · In Progress